

# Project Planning

Project Planning Phase · AI/ML and GenAI Track

Work is divided into sprints following the phases of the template. Replace the [dates] placeholders with your actual schedule.

| Sprint | Focus                   | Deliverables                                                 | Duration |
|--------|-------------------------|--------------------------------------------------------------|----------|
| 1      | Ideation & requirements | Problem statement, empathy map, journey map, DFD, tech stack | [dates]  |
| 2      | Design                  | Problem-solution fit, proposed solution, architecture        | [dates]  |
| 3      | Core development        | Form, validation, Gemini plan generation, database           | [dates]  |
| 4      | Enhancements            | Feedback update, nutrition tip, admin page, error handling   | [dates]  |
| 5      | Testing & hardening     | Functional and performance tests, bug fixes, Docker          | [dates]  |
| 6      | Documentation & demo    | README, project documents, demo rehearsal                    | [dates]  |

## Roles

| Role                | Responsibility                               | Owner  |
|---------------------|----------------------------------------------|--------|
| Project lead        | Planning, coordination, final review         | [Name] |
| Backend developer   | FastAPI routes, database, Gemini integration | [Name] |
| Frontend developer  | Templates, styling, JavaScript               | [Name] |
| Tester / documenter | Test cases, documents, demo preparation      | [Name] |

## Risks and mitigation

| Risk                               | Impact                    | Mitigation                                                       |
|------------------------------------|---------------------------|------------------------------------------------------------------|
| Gemini API rate limits or downtime | Plan cannot be generated  | Retry with backoff; friendly error page; configurable model      |
| Unsafe or low-quality AI advice    | User harm / loss of trust | Safety rules in prompts; medical-consult reminder; feedback loop |
| API key leakage                    | Cost and abuse            | Keep .env out of Git and shared files; rotate keys if exposed    |
| Unprotected admin routes           | Data exposure             | Apply the require_admin dependency to admin routes               |
| Prompt injection via feedback      | Off-topic output          | Feedback is treated as data between markers                      |